Chapter 10: Trigonometric identities and equations

Chapter 10 · Original slide 1 · Trigonometric identities and equations

Y1 Chapter 10
Trigonometric Identities & Equations

Chapter 10 · Original slide 2 · Chapter Overview

1: Know exact trig values for $30^\circ, 45^\circ, 60^\circ$ and understand unit circle.

2: Use identities $\frac{\sin x}{\cos x} \equiv \tan x$ and $\sin^2 x + \cos^2 x \equiv 1$

Show that $3\sin^2 x + 7\sin x = \cos^2 x - 4$ can be written in the form $4\sin^2 x + 7\sin x + 3 = 0$

3: Solve equations of the form $\sin(n\theta) = k$ and $\sin(\theta \pm \alpha) = k$

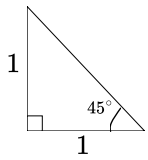
4: Solve equations which are quadratic in sin/cos/tan.

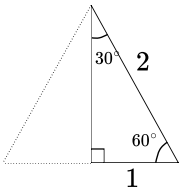
Solve $\sin(2x)(\cos 2x + 1) = 0$, for $0 \leq x < 360^\circ$.

Solve, for $0 \leq x < 360^\circ$, the equation $4\sin^2 x + 9\cos x - 6 = 0$

Chapter 10: Trigonometric identities and equations

Chapter 10 · Original slide 3 · Trigonometric identities and equations





You will frequently encounter angles of 30°, 60°, 45° in geometric problems. Why?

Although you will always have a calculator, you need to know how to derive these.

All you need to remember:

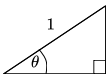
! Draw half a unit square and half an equilateral triangle of side 2.

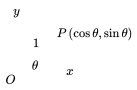
$\sin(30^\circ) =$
$\cos(30^\circ) =$
$\tan(30^\circ) =$
$\sin(60^\circ) =$
$\cos(60^\circ) =$
$\tan(60^\circ) =$

$\sin(45^\circ) =$
$\cos(45^\circ) =$
$\tan(45^\circ) =$

Chapter 10 · Original slide 4 · The Unit Circle and Trigonometry

For values of θ in the range $0 < \theta < 90^\circ$, you know that $\sin\theta$ and $\cos\theta$ are lengths on a right-angled triangle:





But how do we get the rest of the graph for $\sin$, $\cos$ and $\tan$ when $90^\circ \leq \theta \leq 360^\circ$?

! The point P on a unit circle, such that OP makes an angle θ with the positive x-axis, has coordinates $(\cos\theta, \sin\theta)$.
OPhas gradient $\tan\theta$.

And what would be the gradient of the bold line?

Angles are always measured anticlockwise.
(Further Mathematicians will encounter the same when they get to Complex Numbers)

We can consider the coordinate $(\cos\theta, \sin\theta)$ as θ increases from 0 to 360°...

Chapter 10: Trigonometric identities and equations

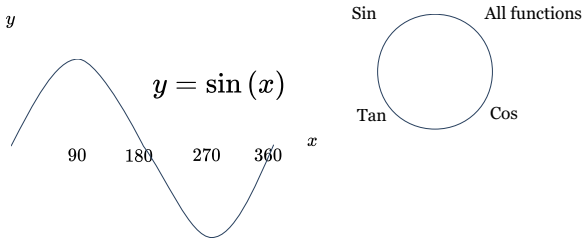
Chapter 10 · Original slide 5 · Mini-Exercise

	$\cos\theta$	$\sin\theta$	$\tan\theta$
$\theta = 0$	1	0	0
$0 < \theta < 90^\circ$			
$\theta = 90^\circ$			
$90^\circ < \theta < 180^\circ$			
$\theta = 180^\circ$			
$180^\circ < \theta < 270^\circ$			
$\theta = 270^\circ$			
$270^\circ < \theta < 360^\circ$			

Use the unit circle to determine each value in the table, using either “0”, “+ve”, “-ve”, “1”, “-1” or “undefined”. Recall that the point on the unit circle has coordinate $(\cos\theta, \sin\theta)$ and OP has gradient $\tan\theta$.

Gradient of OP.

Chapter 10 · Original slide 6 · The Unit Circle and Trigonometry



positive
for $0 < x < 180^\circ$

negative
for $180^\circ < x < 360^\circ$

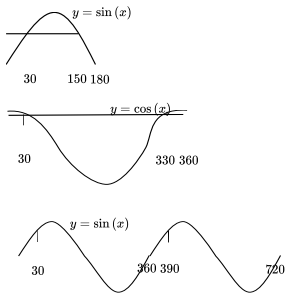
The unit circles explains the behaviour of these trigonometric graphs beyond 90°.

However, the easiest way to remember whether $\sin(x), \cos(x), \tan(x)$ are positive or negative is to just do a very quick sketch (preferably mentally!) of the corresponding graph.

Note: The textbook uses something called ‘*CAST diagrams*’, which is an acronym for which quadrants the Cosine, All, Sine and Tangent are positive

Chapter 10: Trigonometric identities and equations

Chapter 10 · Original slide 7 · A Few Trigonometric Angle Laws



The following are all easily derivable using a quick sketch of a trigonometric graph, and are merely a convenience so you don't always have to draw out a graph every time.

You are highly encouraged to memorise these so that you can do exam questions faster.

1

$\sin(x) = \sin(180^\circ - x)$

e.g. $\sin(150^\circ) = \sin(30^\circ)$

e.g. $\cos(330^\circ) = \cos(30^\circ)$

e.g. $\sin(390^\circ) = \sin(30^\circ)$

Remember from the previous chapter that "cosine" by definition is the sine of the "complementary" angle.

This was/is never covered in the textbook but caught everyone by surprise when it came up in a C3 exam.

We saw this in the previous chapter when covering the 'ambiguous case' when using the sine rule.

2

$\cos(x) = \cos(360^\circ - x)$

3

sin and cos repeat every 360°

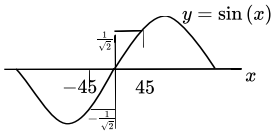
but tan every 180°

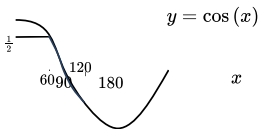
4

$\sin(x) = \cos(90^\circ - x)$

e.g. $\sin(50^\circ) = \cos(40^\circ)$

Chapter 10 · Original slide 8 · Examples





Without a calculator, work out the value of each below.

tan repeats every 180° so can subtract 180°

$\tan(225^\circ) =$

$\tan(210^\circ) =$

$\sin(150^\circ) =$

$\cos(300^\circ) =$

$\sin(-45^\circ) =$

Use the 'laws' where you can, but otherwise just draw out a quick sketch of the graph.

- $\sin(x) = \sin(180 - x)$
- $\cos(x) = \cos(360 - x)$
- sin, cos repeat every 360° but tan every 180°

For sin we can subtract from 180°.

Chapter 10 · Original slide 9 · Test Your Understanding

Without a calculator, work out the value of each below.

$\cos(315^\circ) =$

$\sin(420^\circ) =$

$\tan(-120^\circ) =$

$\tan(-45^\circ) =$

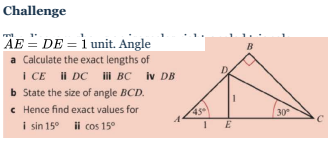
$\sin(135^\circ) =$

- $\sin(x) = \sin(180 - x)$
- $\cos(x) = \cos(360 - x)$
- $\sin, \cos$ repeat every 360° but $\tan$ every 180°

Chapter 10 · Original slide 10 · Exercise 10B – Y1 Page 209

1 Express the following as trigonometric ratios of either 30°, 45° or 60°, *and hence find their exact values.* **a** $\sin 135^\circ$ **b** $\sin(-60^\circ)$ **c** $\sin 330^\circ$ **d** $\sin 420^\circ$ **e** $\sin(-300^\circ)$ **f** $\cos 120^\circ$ **g** $\cos 300^\circ$ **h** $\cos 225^\circ$ **i** $\cos(-210^\circ)$ **j** $\cos 495^\circ$ **k** $\tan 135^\circ$ **l** $\tan(-225^\circ)$ **m** $\tan 210^\circ$ **n** $\tan 300^\circ$ **o** $\tan(-120^\circ)$.

Chapter 10 · Original slide 11 · Exercise 10A/B – Y1 Page 207, 209



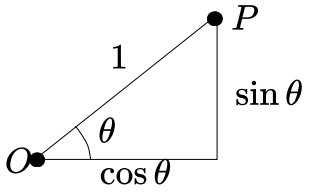
a Calculate the exact lengths of

i CE **ii** DC **iii** BC **iv** DB **b** State the size of angle BCD.

c Hence find exact values for

i $\sin 15°$ **ii** $\cos 15°$

Chapter 10 · Original slide 12 · Trigonometric Identities



1 2

Pythagoras gives you...

Returning to our point on the unit circle...

$\sin^2\theta$ is a shorthand for $(\sin\theta)^2$. It does NOT mean the sin is being squared – this does not make sense as sin is a function, and not a quantity that we can square!

Chapter 10 · Original slide 13 · Application of identities #1: Proofs

Prove that $1 - \tan\theta \sin\theta \cos\theta \equiv \cos^2\theta$

$\tan x = \frac{\sin(x)}{\cos(x)}$

$\sin^2 x + \cos^2 x = 1$

Tip #1: Turn any tan's into sin's and cos's.

Recall that $\equiv$ means 'equivalent to', and just means the LHS is always equal to the RHS for all values of θ.

From Chapter 7 ('Proofs') we saw that usually the best method is to manipulate one side (e.g. LHS) until we get to the other (RHS).

Chapter 10 · Original slide 14 · More Examples

Prove that $\tan\theta + \frac{1}{\tan\theta} \equiv \frac{1}{\sin\theta\cos\theta}$

Tip #3: Look out for $1 - \sin^2\theta$ and $1 - \cos^2\theta$.

Students often don't spot that these can be simplified.

Simplify $5 - 5\sin^2\theta$

Tip #2: In any addition/subtraction involving at least one fraction (with trig functions), always combine algebraically into one.

Chapter 10: Trigonometric identities and equations

Chapter 10 · Original slide 15 · Test Your Understanding

Prove that $\frac{\tan x \cos x}{\sqrt{1-\cos^2 x}} \equiv 1$

Prove that $\frac{\cos^4\theta - \sin^4\theta}{\cos^2\theta} \equiv 1 - \tan^2\theta$

Prove that $\tan^2\theta \equiv \frac{1}{\cos^2\theta} - 1$

Chapter 10 · Original slide 16 · Exercise 10C – Y1 Page 211-212

10 In each of the following, eliminate θ to give an equation relating x and y:

a $x = \sin\theta, y = \cos\theta$ **b** $x = \sin\theta, y = 2\cos\theta$

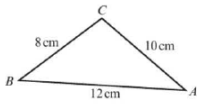
c $x = \sin\theta, y = \cos^2\theta$ **d** $x = \sin\theta, y = \tan\theta$

Problem-solving

In part e find expressi

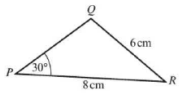
Chapter 10: Trigonometric identities and equations

Chapter 10 · Original slide 17 · Exercise 10C – Y1 Page 211-212



11 The diagram shows the triangle ABC with $AB = 12\,\text{cm}$, $BC = 8\,\text{cm}$ and $AC = 10\,\text{cm}$.

a Show that $\cos B = \frac{9}{16}$. (3 ma



b Hence find the exact value of $\sin B$.

12 The diagram shows triangle PQR with $PR = 8\,\text{cm}$, $QR = 6\,\text{cm}$ and angle $QPR = 30°$.

a Show that $\sin Q = \frac{2}{3}$.

b Given that Q is obtuse, find the exact value of $\cos Q$.

Chapter 10 · Original slide 18 · Extension

[MAT 2008 1C] The simultaneous equations in x, y,

$(\cos\theta)x - (\sin\theta)y = 2$
$(\sin\theta)x + (\cos\theta)y = 1$

are solvable:

A) for all values of θ in range
$0 \leq \theta < 2\pi$

B) except for one value of θ in range $0 \leq \theta < 2\pi$

C) except for two values of θ in range $0 \leq \theta < 2\pi$

D) except for three values of θ in range $0 \leq \theta < 2\pi$

Chapter 10 · Original slide 19 · Solving Trigonometric Equations

Remember those trigonometric angle laws (on the right) earlier this chapter? They're about to become very useful!

Reminder of 'trig laws':

- $\sin(x) = \sin(180 - x)$
- $\cos(x) = \cos(360 - x)$
- $\sin, \cos$ repeat every 360° but $\tan$ every 180°

Solve $\sin\theta = \frac{1}{2}$ in the interval $0 \leq \theta \leq 360^\circ$.

Calculator Note:

When you do sin^{-1}, cos^{-1} and tan^{-1} on a calculator, it gives you only one value, known as the principal value.

Solve $5\tan\theta = 10$ in the interval $-180^\circ \leq \theta < 180^\circ$

Tip: Look out for the solution range required. $-180 \leq \theta < 180^\circ$ is a particularly common one.

$\tan$ repeats every 180°, so can add/subtract 180° as we please.

Chapter 10 · Original slide 20 · Slightly Harder Ones…

Solve $\sin\theta = -\frac{1}{2}$ in the interval $0 \leq \theta \leq 360^\circ$.

In general you should have 2 solutions per 360° (except when at a peak or trough of the trig graph)

Note that we've had to use a second law, i.e. that $\sin$ repeats every 360°.

Solve $\sin\theta = \sqrt{3}\cos\theta$ in the interval $0 \leq \theta \leq 360^\circ$.

Hint: The problem here is that we have two different trig functions. Is there anything we can divide both sides by so we only have one trig function?

Chapter 10 · Original slide 21 · Test Your Understanding

Solve $2\cos\theta = \sqrt{3}$ in the interval $0 \leq \theta \leq 360°$.

Solve $\sqrt{3}\sin\theta = \cos\theta$ in the interval $-180° \leq \theta \leq 180°$.

Chapter 10 · Original slide 22 · Exercise 10D – Y1 215-216

7 A teacher asks two students to solve the equation $2\cos x = 3\sin x$ for $-180° \leq x \leq 180°$. The attempts are shown:
Student A: $\tan x = \frac{3}{2}$, $x = 56.3°$ or $x = -123.7°$.
Student B: $4\cos^2 x = 9\sin^2 x$,
$4(1 - \sin^2 x) = 9\sin^2 x$, $4 = 13\sin^2 x$,
$\sin x = \pm\sqrt{\frac{4}{13}}$, $x = \pm 33.7°$ or $x = \pm 146.3°$.
a Identify the mistake made by Student A. (1 mark)
b Identify the mistake made by Student B and explain the effect it has on their solution. (2 marks)
c Write down the correct answers to the question. (1 mark)

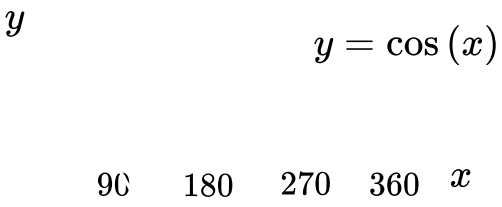
9 Find all the values of θ, to 1 decimal place, in the interval $0 < \theta < 360°$ for which $\tan^2\theta = 9$

Chapter 10 · Original slide 23 · Exercise 10D – Y1 215-216

10 a Show that $4\sin^2 x - 3\cos^2 x = 2$ can be written as $7\sin^2 x = 5$.
b Hence solve, for $0 \leq x \leq 360°$, the equation $4\sin^2 x - 3\cos^2 x = 2$. Give your answers to 1 decimal place.

11 a Show that the equation $2\sin^2 x + 5\cos^2 x = 1$ can be written as $3\sin^2 x = 4$.
b Use your result in part a to explain why the equation $2\sin^2 x + 5\cos^2 x = 1$ has no solutions.

Chapter 10 · Original slide 24 · Harder Equations



Harder questions replace the angle θ with a linear expression.

Solve $\cos 3x = -\frac{1}{2}$ in the interval $0 \leq x \leq 360°$.

STEP 1: Adjust the range of values for θ to match the expression inside the cos.

STEP 2: Immediately after applying an inverse trig function (and BEFORE dividing by 3!), find all solutions up to the end of the interval.

In general you tend to get a pair of values per $360°$ (for any of sin/cos/tan), except for $\cos\theta = \pm 1$ or $\sin\theta = \pm 1$:

Thus once getting your first pair of values (e.g. using $\sin(180 - \theta)$ or $\cos(360 - \theta)$ to get the second value), keep adding $360°$ to generate new pairs.

STEP 3: Then do final manipulation to each value.

Chapter 10 · Original slide 25 · Further Examples

Solve $\sin(2x + 30^\circ) = \frac{1}{\sqrt{2}}$ in the interval $0 \leq x \leq 360^\circ$.

To get from x to $2x + 30^\circ$ we double and add 30°. So do the same to the upper and lower bound!

Solve $\sin x = 2\cos x$ in the interval $0 \leq x < 300^\circ$

By dividing both sides by $\cos x$, the $\sin x$ becomes $\tan x$ and the $\cos x$ disappears, leaving a trig equation helpfully only in terms of one trig function.

Chapter 10 · Original slide 26 · Test Your Understanding

Solve, for $0 \leq x < 180^\circ$,
$\cos(3x - 10^\circ) = -0.4$,
giving your answers to 1 decimal place. You should show each step in your working. (7)

Chapter 10 · Original slide 27 · Exercise 10E – Y1 p219

4 Solve for $0 \leq x \leq 180^\circ$ the equations:
a $\sin(x + 20^\circ) = \frac{1}{2}$.
b $\cos 2x = -0.8$, giving your answers to 1 decimal place.

5 a Sketch for $0 \leq x \leq 360^\circ$ the graph of $y = \sin(x + 60^\circ)$.
b Write down the exact coordinates of the points where the graph meets the coordinate axes.
c Solve, for $0 \leq x \leq 360^\circ$, the equation $\sin(x + 60^\circ) = 0.55$, giving your answers to 1 decimal place.

Chapter 10 · Original slide 28 · Exercise 10E – Y1 p219

6 a Given that $4 \sin x = 3 \cos x$, write down the value of $\tan x$.
b Solve, for $0 \leq \theta \leq 360^\circ$, $4 \sin 2\theta = 3 \cos 2\theta$ giving your answers to 1 decimal place.

7 The equation $\tan kx = -\frac{1}{\sqrt{3}}$, where k is a constant and $k > 0$, has a solution at $x = 60^\circ$.
a Find a possible value of k.
b State, with justification, whether this is the only such possible value of k.

Chapter 10 · Original slide 29 · Quadratics in sin/cos/tan

We saw that an equation can be ‘quadratic in’ something, e.g. $x - 2\sqrt{x} + 1 = 0$ is ‘quadratic in $\sqrt{x}$’, meaning that $\sqrt{x}$ could be replaced with another variable, say y, to produce a quadratic equation $y^2 - 2y + 1 = 0$.

Solve $5\sin^2 x + 3\sin x - 2 = 0$ in the interval $0 \leq x \leq 360^\circ$.

Method 1: Use a substitution.

Method 2: Factorise without substitution.

This is the same, but we ‘imagine’ $\sin x$ as a single variable and hence factorise immediately.

Chapter 10 · Original slide 30 · More Examples

Solve $\tan^2\theta = 4$ in the interval $0 \leq x \leq 360^\circ$.

Missing the negative case would result in the loss of multiple marks. Beware!

Solve $2\cos^2 x + \sin x = 3\sin^2 x$ in the interval $-180^\circ \leq x \leq 180^\circ$.

Tip: We have an identity involving sin^2 and cos^2, so it makes sense to change the squared one that would match all the others.

Chapter 10 · Original slide 31 · Test Your Understanding

(a) Show that the equation $5\sin x = 1 + 2\cos^2 x$ can be written in the form $2\sin^2 x + 5\sin x - 3 = 0$. (2)
(b) Solve, for $0 \leq x < 360^\circ$, $2\sin^2 x + 5\sin x - 3 = 0$. (4)

Chapter 10 · Original slide 32 · Exercise 10F – Y1 P221-222

5 Find all the solutions, in the interval $0 \leq x \leq 360^\circ$, to the equation $8\sin^2 x + 6\cos x - 9 = 0$ giving each solution to one decimal place. (6 marks)

6 Find, for $0 \leq x \leq 360^\circ$, all the solutions of $\sin^2 x + 1 = \frac{7}{2}\cos^2 x$ giving each solution to one decimal place.

Chapter 10 · Original slide 33 · Exercise 10F – Y1 P221-222

7 Show that the equation $2\cos^2 x + \cos x - 6 = 0$ has no solutions.

b Hence show that the equation $\cos^2 x = 2 - \sin x$ has no solutions. (3 marks)

8 a Show that the equation $\cos^2 x = 2 - \sin x$ can be written as $\sin^2 x - \sin x + 1 = 0$. (2 marks)

Chapter 10 · Original slide 34 · Exercise 10F – Y1 P221-222

9 $\tan^2 x - 2\tan x - 4 = 0$.
a Show that $\tan x = p \pm \sqrt{q}$ where p and q are numbers to be found.
b Hence solve the equation $\tan^2 x - 2\tan x - 4 = 0$ in the interval $0 \leq x \leq 540°$.

Chapter 10: Trigonometric identities and equations

Chapter 10 · Original slide 35 · Extension

1

[MAT 2010 1C] In the range $0 \leq x < 360^\circ$, the equation

$$\sin^2 x + 3\sin x \cos x + 2\cos^2 x = 0$$

Has how many solutions?

2

[MAT 2014 1E] As x varies over the real numbers, the largest value taken by the function

$(4\sin^2 x + 4\cos x + 1)^2$ equals what?